Pandas Cheatsheet — 40 Essential Functions

Category	Function	Description
Data Loading & Inspection	`pd.read_csv("file.csv")`	Load CSV
	`pd.read_json("data.json")`	Load JSON
	`df.info()`	Schema + nulls
	`df.head(10)`	Preview rows
	`df.sample(5)`	Random sample
	`df.shape`	Rows/columns
	`df.describe()`	Summary stats
Selecting & Filtering	`df['col']`	Select column
	`df[['a','b']]`	Select multiple columns
	`df[df['age'] > 30]`	Boolean filter
	`df[(df['age']>18) & (df['country']=='US')]`	Multi-condition
	`df.query("country=='US'")`	SQL-style filter
	`df[df['age'].between(20,30)]`	Range filter
Cleaning & Transforming	`df.dropna(subset=['id'])`	Drop missing
	`df.fillna(0)`	Fill missing
	`df.drop_duplicates(subset=['id'])`	Deduplicate
	`df.duplicated()`	Check duplicates
	`df.columns.difference(['id'])`	Exclude columns
	`df.reset_index(drop=True)`	Reset index
	`df.rename(columns={'a':'A'})`	Rename
	`df.drop(columns=['col'])`	Drop column
	`df.replace({'A':1})`	Replace values
	`np.where(df['x']>0,1,0)`	Conditional column
Type Handling	`df['id'].astype('Int64')`	Cast type
	`pd.to_datetime(df['ts'])`	To datetime
	`df['ts'].dt.date`	Extract date
	`df['ts'].dt.year`	Extract year
Aggregation & Grouping	`df.groupby('cat')['sales'].sum()`	Group + sum
	`df.groupby('id').agg({'a':'sum','b':'count'})`	Multi-agg
	`df['id'].count()`	Count
	`df['user_id'].nunique()`	Unique count
	`df['country'].value_counts()`	Frequency
	`df.sort_values('sales', ascending=False)`	Sort
Joining & Combining	`pd.merge(a, b, on='id', how='left')`	SQL join
	`df1.join(df2, how='left')`	Index join
	`pd.concat([df1, df2])`	Stack DataFrames
Advanced Operations	`df['ma7'] = df['sales'].rolling(7).mean()`	Rolling window
	`df['r'] = df.groupby('cat')['sales'].rank()`	Rank
	`df.pivot_table(values='sales', index='cat', columns='region')`	Pivot
	`df['x2'] = df['x'].apply(lambda x: x*2)`	Apply
	`df['email'].str.lower()`	String ops
	`df.loc[df['age']<0, 'age'] = 0`	Update rows

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